

Oracle VirtualBox

Oracle VirtualBox is a Type2 hypervisor that can run on Linux, Windows, Macintosh and Solaris hosts. It is portable as it can run on a large number of 32-bit and 64-bit host operating systems. It's called a hosted hypervisor as it requires an existing operating system to be installed. One good feature of VirtualBox is that virtual machines can be easily imported and exported using OVF (Open Virtualization Format). It's even possible to import OVFs that are created by different virtualisation software. The detailed feature list is available at <http://download.virtualbox.org/virtualbox/UserManual.pdf>.

VMware Workstation Player

VMware Workstation Player is a Type2 desktop virtualisation application that provides a streamlined user interface for running and evaluating operating systems and applications in a virtual machine on Windows or Linux machines. Its simple UI makes it the easiest way to deliver a virtual desktop to your employees, contractors, or customers. It's available for download from https://my.vmware.com/web/vmware/free#desktop_end_user_computing/vmware_workstation_player/12_0.

Lguest

Lguest is a very lightweight hypervisor built into the Linux kernel. The core of Lguest is the driver module (named 'lg') available from Linux kernel 2.6.23 and above. Lguest provides para-virtualised solutions for Linux. The 'lg' driver module, during its initialisation, allocates a chunk of memory and maps to the kernel's address space and a small hypervisor is loaded into this memory area. It also provides a virtualised I/O subsystem. That being said, it doesn't provide any of the fancy features other hypervisors do; however, it is a good option when you need to develop and test the kernel boot. To play around with Lguest, users can take a look at the LWN article^[9]. The documentation and download details are found at <http://lguest.ozlabs.org/>.

Linux Containers (LXC)/Docker

Linux Containers (LXC) is an operating system-level

virtualisation method for running multiple isolated Linux systems (containers) on a single control host (LXC host). It does not provide a virtual machine, but a virtual environment that has its own CPU, memory, block I/O, network, etc. It makes use of the Linux kernel's cgroups functionality, which provides isolated namespace(s) to run isolated applications. One advantage of containers is that they don't require a full-fledged guest OS like virtual machines. Refer to the site^[10] to understand more about containers.

Linux-VServer

Linux-VServer is an OS level virtualisation solution. It's a soft partitioning concept technology based on security contexts. It basically creates virtual private servers (VPS) that run simultaneously on a single physical server by sharing hardware resources. Each VPS has its own database account and root password so that it's isolated from other virtual servers. Refer to the Linux-VServer website^[11] to understand the solution better and to download it. 

References

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